PulmoTwin: Outpatient Respiratory Digital Twin

Slide 1: Title Slide

A 48-Hour Predictive Early Warning System for COPD & Asthma Exacerbations

- Team: PulmoTwin Engineering Group
- Competition: Happiest Health Digital Twin Challenge 2026
- Modality: Static EHR + Continuous Wearables + Open-Meteo CAMS Air Quality
- Open Source: MIT License | Decision-Support Research Prototype

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The High-Cost Crisis of Respiratory Exacerbations

Slide 2: Clinical Problem Statement

- Clinical Reality: COPD and severe asthma affect >500 million people globally, causing >3.2 million annual deaths.
- The Critical Window: Acute flare-ups are treated reactively in emergency departments when airways have already failed.
- Physiological Drift: Airway inflammation begins 48–72 hours prior to severe symptoms (micro-drops in SpO2, tachycardia, mobility declines).
- Environmental Catalysts: Delhi NCR's severe winter smog (AQI > 400) and spring tree pollen trigger acute hospitalization cascades.
- The Gap: No current clinical tool fuses personal physiological baselines with real-time atmospheric exposure for advance warning.

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PulmoTwin: Patient-Specific Predictive Outpatient Twin

Slide 3: Proposed Healthcare Digital Twin

- Dynamic In Silico Twin: Maintains longitudinal personal baseline for SpO2, heart rate, respiration, sleep, and daily steps.
- Multi-Modal Fusion: Automatically aligns continuous wearable telemetry with local CAMS atmospheric air quality and ERA5 weather.
- 48-Hour Advance Warning: Predicts impending acute exacerbations before severe respiratory decompensation occurs.
- Counterfactual Simulation ('What-If'): Allows clinicians to quantify the precise risk reduction of air filtration and medication compliance.
- Workload Feasible: Designed for hospital nurse navigators with controlled alert burden (0.77 alerts/patient/month).

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365-Day Cohort & Multi-Modal Dataset Architecture

Slide 4: Data Engineering & Provenance

- Synthetic Static Cohort (N=500): Clinically grounded COPD & asthma cohort (age 42–84, sex, smoking pack-years, staging, adherence).
- Longitudinal Vitals (182,500 patient-days): 365 daily records per patient with physiological drift modeled on published literature.
- Real Delhi Air Quality: Hourly CAMS data from 5 Delhi stations (Mandir Marg, Anand Vihar, Punjabi Bagh, RK Puram, IGI Airport).
- Weather & Dust (ERA5): Daily mean temperature, diurnal swing (ΔT), wind speed, precipitation, and atmospheric dust.
- Delhi Seasonal Pollen Model: Bimodal aerobiological curve calibrated to AIIMS & VPCI literature (tree & grass pollen peaks).
- Strict Compliance: Synthetic, anonymized, and open data only. No real patient health information (PHI) utilized.

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XGBoost Classifier Benchmarking Across 4 Modalities

Slide 5: Machine Learning & Ablation Study

- Train/Test Strategy: Strict patient-level 80/20 split (400 train / 100 test patients; 0 data leakage).
- Evaluation Metrics (Stage 4 Multi-Modal):
  - ROC-AUC: 0.9877 | PR-AUC: 0.9080
  - Test Sensitivity (Recall): 93.4% @ optimal decision threshold
  - Test Precision (PPV): 67.8% | F1-Score: 0.7857
- 4-Stage Ablation Progression:
  - M1 (Wearable Vitals Only): ROC-AUC 0.9412 | PR-AUC 0.7854
  - M2 (+ Static EHR Staging & Smoking): ROC-AUC 0.9634 | PR-AUC 0.8320
  - M3 (+ CAMS AQI & PM2.5 Exposure): ROC-AUC 0.9782 | PR-AUC 0.8745
  - M4 (+ Full Multi-Modal Weather, Dust, Pollen): ROC-AUC 0.9877 | PR-AUC 0.9080 (+12.3% PR-AUC gain)

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Clinical Operational Feasibility Analysis

Slide 6: Lead-Time Warning & Workload

- Warning Horizon Sensitivity:
  - 48 Hours Ahead: 93.4% Sensitivity (Recall)
  - 36 Hours Ahead: 94.2% Sensitivity
  - 24 Hours Ahead: 95.8% Sensitivity
- Clinical Workload & Workload Feasibility:
  - Average Alerts per Patient per Month: 0.77 alerts
  - Consecutive Alert Days Grouped: Multi-day alerts combined into single clinical episodes (e.g. '8–9 Jan, 2-day episode').
- Sensor Dropout Robustness Test:
  - At 10% wearable data dropout: PR-AUC remains 0.892 (98.2% retention).
  - At 30% wearable data dropout: PR-AUC remains 0.854 (94.1% retention).

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Clinically Transparent TreeExplainer Attribution

Slide 7: Explainable AI & SHAP Decomposition

- TreeExplainer Log-Odds to Sigmoid Probability Attribution:
  - Shared percentage points mathematically reconcile: Baseline (3.6%) + Net Bars = Displayed Risk%.
- 9 Actionable Clinical Factors:
  1. SpO2 vs personal baseline (primary physiological trigger)
  2. Resting heart rate elevation (sub-clinical tachycardia)
  3. Respiratory rate acceleration (tachypnea)
  4. Sleep duration & nocturnal fragmentation
  5. Daily mobility reduction (step count collapse)
  6. CAMS NAQI exposure (ambient Delhi smog)
  7. Atmospheric dust concentration
  8. Weather & seasonal pollen index
  9. Inhaler adherence history

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Hospital-Grade Monitoring Interface

Slide 8: Interactive React Clinical Dashboard

- Editorial-Clinical Aesthetic: Calm warm surface, deep teal accent, large hero risk numeral, zero visual clutter.
- 3D Respiratory Digital Twin: Interactive Three.js lung model breathing at patient's respiratory rate, tinted teal->amber->coral.
- 365-Day Live Simulation Engine: Play/Pause/Speed controls, date scrubber, and 'Jump to Next Alert' button.
- Dark 'Patient Monitor' Telemetry: Deep navy card, glowing teal SpO2, soft coral HR, amber CAMS NAQI, and steps bars.
- 5 Vital Tiles: SpO2, HR, RR, Sleep, Steps with 7-day sparklines and baseline deviation indicators.
- Interactive Delhi GIS Map: 5 ambient stations with patient pinned and local micro-climate readings.
- 1-Click Nurse Actions: Dispatch oral steroids, arrange telemedicine triage, or deliver air purifier units.

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Simulating Outpatient Preventive Interventions

Slide 9: Counterfactual Digital Twin ('What-If')

- Interventions Simulated in Real-Time:
  - Indoor HEPA Air Filtration (AQI reduction: 0 to -150 points)
  - Pharmacotherapy Adherence Support (+0% to +50% compliance)
  - Indoor Thermal Swing Stabilization (0 to -8°C swing dampening)
- Clinical Impact on High-Risk Days:
  - On high-risk patients (e.g. PT-265 at 63.6% baseline risk), HEPA air cleaning + inhaler support drops risk by 42.2–56.0 points down to 7.6% Low Risk (88.1% rel
  - On stable days (~0.1% baseline floor), verifies that protective regimens maintain ongoing stability.

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Project Deliverables & Public Access Guarantee

Slide 10: Summary & Submission Verification

- Repository Stack: Python 3.10+ (FastAPI, XGBoost, SHAP, Pandas, Scikit-Learn) + React 18 (Vite, Lucide, Three.js).
- Full Documentation: /docs/architecture.md, /docs/deck_outline.md, and comprehensive README.md.
- Vector PDF Deliverables in /docs:
  - docs/PulmoTwin_Architecture_Diagram.pdf (Vector multi-layer pipeline)
  - docs/PulmoTwin_Presentation_Deck.pdf (10-slide executive pitch deck)
- Video Walkthrough: [Unlisted YouTube Video Demo Link](YOUR_YOUTUBE_LINK_HERE)
- Open Source: MIT License | Clean working tree under GitHub 100MB limit (dataset: 62.98 MB).
- Disclaimer: Decision-support research prototype. Not a diagnostic medical device.

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